EXPERIMENT - 73

Question 1

1. Problem Statement

Create an enumeration for days (SUNDAY to SATURDAY) and print each day with its integer value.
Input: No Input
Output: SUNDAY = 0, MONDAY = 1, TUESDAY = 2, WEDNESDAY = 3, THURSDAY = 4, FRIDAY = 5, SATURDAY = 6

2. Algorithm

Step 1: START
Step 2: Declare enum Day with SUNDAY, MONDAY, TUESDAY, WEDNESDAY, THURSDAY, FRIDAY and SATURDAY, so their values are 0 to 6.
Step 3: Repeat for day = SUNDAY to SATURDAY: display the name of the day, =, and its integer value, with a comma after every day except SATURDAY.
Step 4: STOP

3. Test Case Table

#	Input	Expected Output	Actual Output	Result
1	No Input	SUNDAY = 0, MONDAY = 1, TUESDAY = 2, WEDNESDAY = 3, THURSDAY = 4, FRIDAY = 5, SATURDAY = 6	SUNDAY = 0, MONDAY = 1, TUESDAY = 2, WEDNESDAY = 3, THURSDAY = 4, FRIDAY = 5, SATURDAY = 6	**PASS**

4. Source Code

days73.c

```
/* Experiment 73, Question 1: enumeration for the days of the week. */
#include <stdio.h>

enum Day { SUNDAY, MONDAY, TUESDAY, WEDNESDAY, THURSDAY, FRIDAY, SATURDAY };

int main(void)
{
    const char *names[] = {"SUNDAY", "MONDAY", "TUESDAY", "WEDNESDAY",
                           "THURSDAY", "FRIDAY", "SATURDAY"};
    enum Day day;

    for (day = SUNDAY; day <= SATURDAY; day++) {
        printf("%s = %d", names[day], day);
        if (day < SATURDAY)
            printf(", ");
    }
    printf("\n");
    return 0;
}
```

Question 2

1. Problem Statement

Define an enumeration for traffic lights (RED, YELLOW, GREEN) and print Stop, Wait, or Go based on its value.
Input: GREEN
Output: Go

2. Algorithm

Step 1: START
Step 2: Declare enum Light with RED, YELLOW and GREEN.
Step 3: Read the color and change its letters to capital letters.
Step 4: If it is RED, YELLOW or GREEN, set light to that value. Otherwise display Invalid traffic light color and stop.
Step 5: If light is RED display Stop, if YELLOW display Wait, and if GREEN display Go.
Step 6: STOP

3. Test Case Table

#	Input	Expected Output	Actual Output	Result
1	GREEN	Go	Go	**PASS**
2	RED	Stop	Stop	**PASS**
3	YELLOW	Wait	Wait	**PASS**
4	green	Go	Go	**PASS**
5	BLUE	Invalid traffic light color	Invalid traffic light color	**PASS**

4. Source Code

light73.c

```
/* Experiment 73, Question 2: enumeration for traffic lights. */
#include <stdio.h>
#include <string.h>
#include <ctype.h>

enum Light { RED, YELLOW, GREEN };

int main(void)
{
    char input[20];
    enum Light light;
    int i;

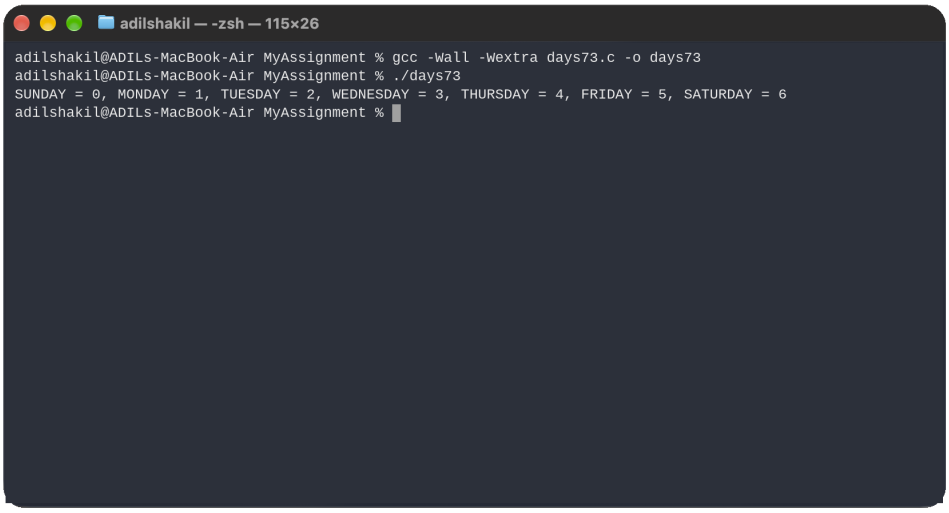
    printf("Enter the traffic light color: ");
    scanf("%19s", input);
    for (i = 0; input[i] != '\0'; i++)
        input[i] = toupper((unsigned char)input[i]);

    if (strcmp(input, "RED") == 0)
        light = RED;
    else if (strcmp(input, "YELLOW") == 0)
        light = YELLOW;
    else if (strcmp(input, "GREEN") == 0)
        light = GREEN;
    else {
        printf("Invalid traffic light color\n");
        return 1;
    }

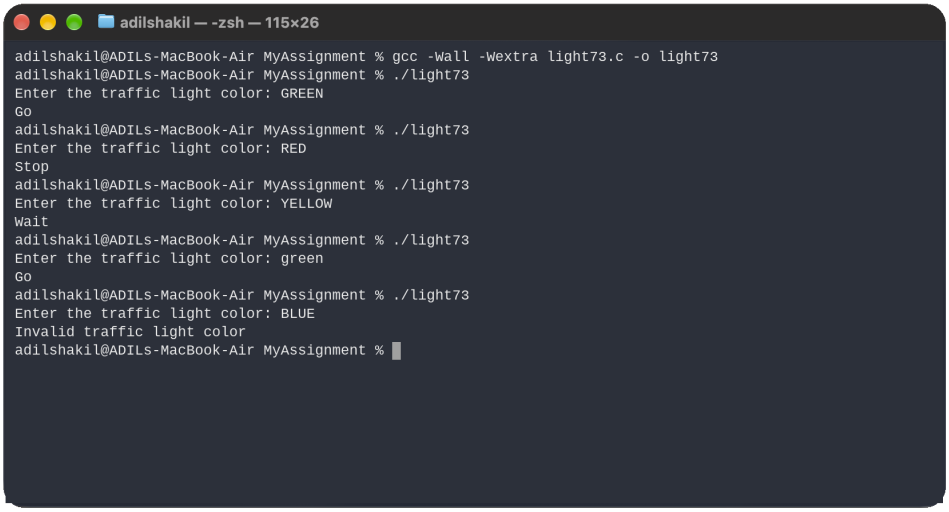
    switch (light) {
    case RED:
        printf("Stop\n");
        break;
    case YELLOW:
        printf("Wait\n");
        break;
    case GREEN:
        printf("Go\n");
        break;
    }
    return 0;
}
```

5. Compilation and Output

Question 1: compiled with `gcc -Wall -Wextra days73.c -o days73`



Question 2: compiled with `gcc -Wall -Wextra light73.c -o light73`



Both programs built clean, and the outputs match the question.